

GlobalTech Financial Data Analytics & Audit

Capstone project

Name: Ayush Kumar

Internship by: iStudio

Course / Batch: Data Analytics Internship (2025-26)

Executive Summary

Split-layout card view with 3 primary metrics highlighted in large bold text on the left, and key takeaways on the right.

Key Metrics Cards:

- \$148.27M USD: Total Audited Revenue
- 14,183: Verified Active Accounts
- < 2.0 Seconds: Full Pipeline Execution Time

Key Takeaways:

- Resolved multi-source data fragmentation across relational SQL, raw server logs, and static exchange rate files.
- Eliminated system error logs and deduplicated user status histories using windowing logic.
- Provided real-time executive visibility via an interactive Power BI audit dashboard.

Target Audience:

4-quadrant grid detailing impact across different corporate teams.

- Executive Leadership & CFO: Delivers an audited, single source of truth for total global USD revenue and quarterly financial performance.
- Financial Auditors & Compliance: Verifies cross-border transaction integrity, exchange rate consistency, and account deduplication accuracy.
- Data Engineering & Analytics: Establishes a scalable, reproducible ETL pipeline architecture for future automated reporting.
- Account Management & Sales Strategy: Provides customer-level contribution metrics to drive enterprise retention and growth.

Problem Statement

3 horizontal risk callout boxes.

- **Unstructured & Fragmented Logs:** Over 100,000 server log entries contained unstructured JSON payloads mixed with system error logs that skewed raw calculations.
- **Account Status Duplication:** Users transition through Pending, Active, and Suspended states over time; without deduplication, suspended accounts risked polluting active figures.
- **Temporal Exchange Rate Gaps:** Currency exchange rate tables omitted weekends and holidays, leaving cross-border transactions unrecorded if not handled programmatically.

Data Quality & Cleaning Metrics Slide

- Raw Log Ingestion: 100,000 raw server log entries processed.
- Error Log Filtration: ~5,098 system error/failed transaction logs identified and removed.
- Valid Parsed Transactions: 94,902 clean financial transaction records.
- Active Deduplicated Transactions: 53,899 reconciled transactions belonging strictly to verified active accounts.

Risk, Governance & Audit Compliance

In financial fin-tech environments, compliance and data security are critical.

- **Auditability & Traceability:** Every line item in the final Power BI dashboard maps back directly to a parsed raw log line and timestamp.
- **Deduplication Safeguards:** Used SQL window functions to ensure suspended or pending users do not artificially inflate company valuation or active user KPIs.
- **Exchange Rate Integrity:** Eliminated missing weekend data points by standardizing rates via Friday forward-fill logic, preventing artificial weekend revenue dips

Technology Stack

Pipeline Phase	Technology / Library
Database Querying	SQLite3
Data Engineering	Python 3.x (Pandas, Regex)
Static Analytics	Matplotlib & Seaborn
BI Reporting	Power BI Desktop

End-to-End ETL Pipeline Methodology

- A 4-step horizontal process flowchart with arrows connecting each phase.
- Phase 1: SQL Extraction → Execute window functions to isolate current Active users (14,183 verified accounts).
- Phase 2: Vectorized Regex Parsing → Apply regex patterns to discard system errors and extract 94,902 valid transaction logs (eur_val).
- Phase 3: Temporal Forward-Fill → Reindex 2023 calendar dates and roll Friday exchange rates forward across weekend transactions.
- Phase 4: USD Revenue Normalization → Multiply Euro values by daily exchange rates to yield 53,899 reconciled active user transactions.

Power BI Executive Dashboard Overview

Full-slide screenshot of the Power BI Dashboard with callout arrows pointing to key visual elements.

Visual Highlights:

- Top Header & Date Slicer: Enables interactive filtering across Q1–Q4 or individual date ranges.
- KPI Summary Cards: Displays Total Revenue (\$148.27M), Total Volume (54K transactions), and Average Ticket Size (\$2.75K).
- Monthly Revenue Trajectory: Demonstrates consistent monthly revenue velocity without major seasonal drops.
- Top 5 Customer Bar Chart: Highlights leading accounts (User_U-06582 at ~\$40.05K) showing balanced revenue distribution.

Customer Contribution & Growth Insights

Side-by-side comparison cards highlighting revenue stability and customer concentration risk.

Revenue Trajectory Analysis:

- Monthly revenue velocity remained steady across Q1–Q4, averaging ~\$12.35M USD per month.
- Confirms strong product-market fit and recurring operational stability without reliance on seasonal spikes.

Customer Concentration Analysis:

- The top 5 enterprise accounts contribute ~\$196K USD combined.
- Top spender User_U-06582 generated \$40,050.99 USD, followed closely by User_U-23313 at \$39,555.81 USD.
- Demonstrates healthy account diversification with low single-customer concentration risk.

Results

✓ Phase 1 Complete: Extracted 14,183 active users.

1 to 5 of 5 entries [Filter](#) [📄](#) [?](#)

index	user_id	name	status	last_updated
0	U-00001	User_U-00001	Active	2023-01-28 14:00:00
1	U-00003	User_U-00003	Active	2023-01-22 14:00:00
2	U-00005	User_U-00005	Active	2023-01-17 14:00:00
3	U-00006	User_U-00006	Active	2023-01-05 14:00:00
4	U-00007	User_U-00007	Active	2023-01-27 14:00:00

Show per page

✓ Phase 2 Complete: Extracted 94,982 valid transaction records.

1 to 5 of 5 entries [Filter](#) [📄](#) [?](#)

index	Date	User_ID	Product_ID	Euro_Value
0	2023-03-15	U-19484	PRD-874	3404.44
1	2023-05-09	U-18932	PRD-905	3406.21
2	2023-10-21	U-15882	PRD-284	164.08
3	2023-01-10	U-22781	PRD-665	26.8
4	2023-05-23	U-13348	PRD-589	4964.77

Show per page

✓ Phase 3 Complete: Processed 53,899 active transactions.
✓ Audited Total Revenue: \$148,274,622.91 USD

1 to 5 of 5 entries [Filter](#) [📄](#) [?](#)

index	Date	User_ID	Product_ID	Euro_Value	EUR_to_USD	name	USD_Revenue
0	2023-01-01 00:00:00	U-14955	PRD-430	2161.06	1.1248357076505615	User_U-14955	2430.8374543753225
1	2023-01-01 00:00:00	U-07290	PRD-963	2892.39	1.1248357076505615	User_U-07290	3253.4635524514074
2	2023-01-01 00:00:00	U-09683	PRD-952	3486.6	1.1248357076505615	User_U-09683	3921.8521782944476
3	2023-01-01 00:00:00	U-19319	PRD-331	2879.4	1.1248357076505615	User_U-19319	2338.9833704885777
4	2023-01-01 00:00:00	U-20903	PRD-785	4161.26	1.1248357076505615	User_U-20903	4680.733836817976

Show per page

Results

```
... --- DATA QUALITY REPORT ---
```

```
1. Missing Values in Final Dataset:
```

```
Date          0
User_ID       0
Product_ID    0
Euro_Value    0
EUR_to_USD    0
name          0
USD_Revenue   0
dtype: int64
```

```
2. Duplicate Rows Count: 0
```

```
3. Date Range Covered: 2023-01-01 to 2023-12-31
```

```
4. Revenue Summary Statistics ($ USD):
```

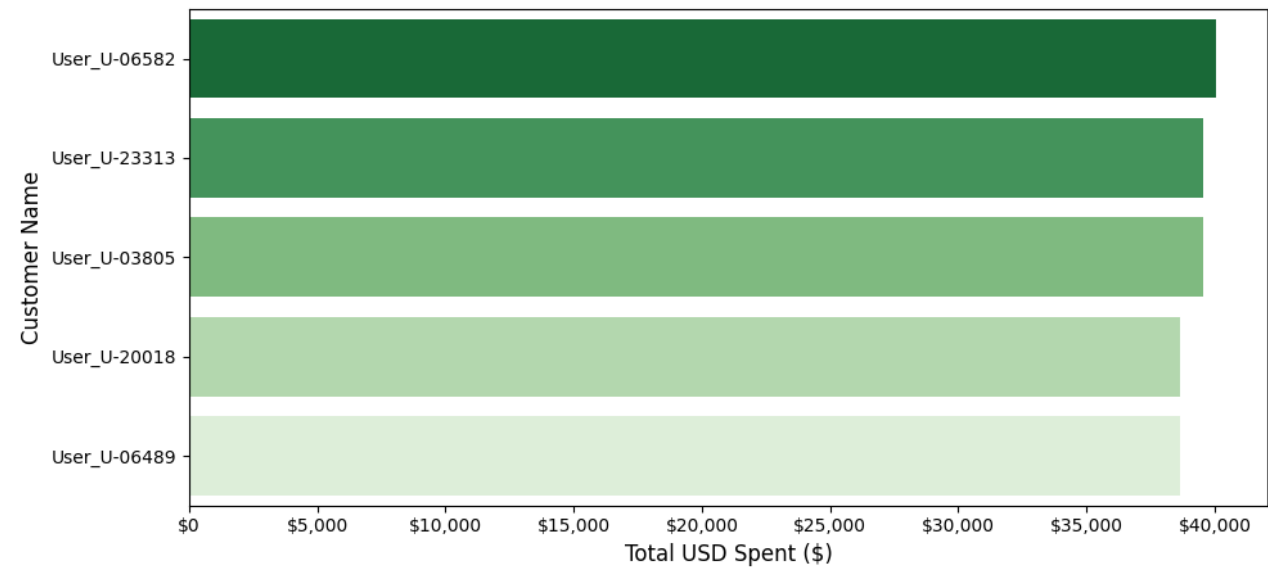
```
    USD_Revenue
```

count	\$53,899.00
mean	\$2,750.97
std	\$1,592.94
min	\$10.61
25%	\$1,365.18
50%	\$2,742.68
75%	\$4,120.10
max	\$6,161.30

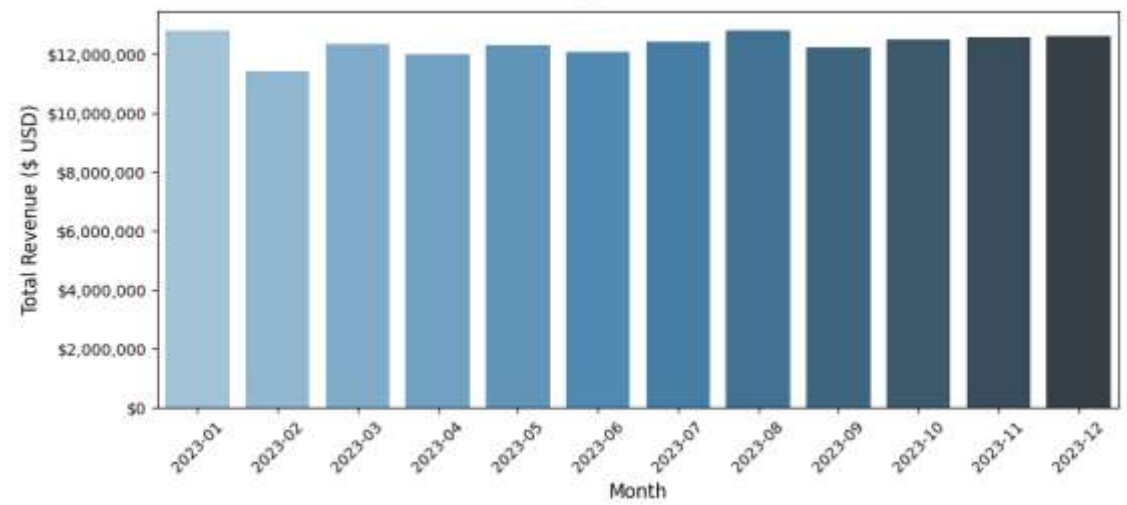
```
dtype: object
```

Results:

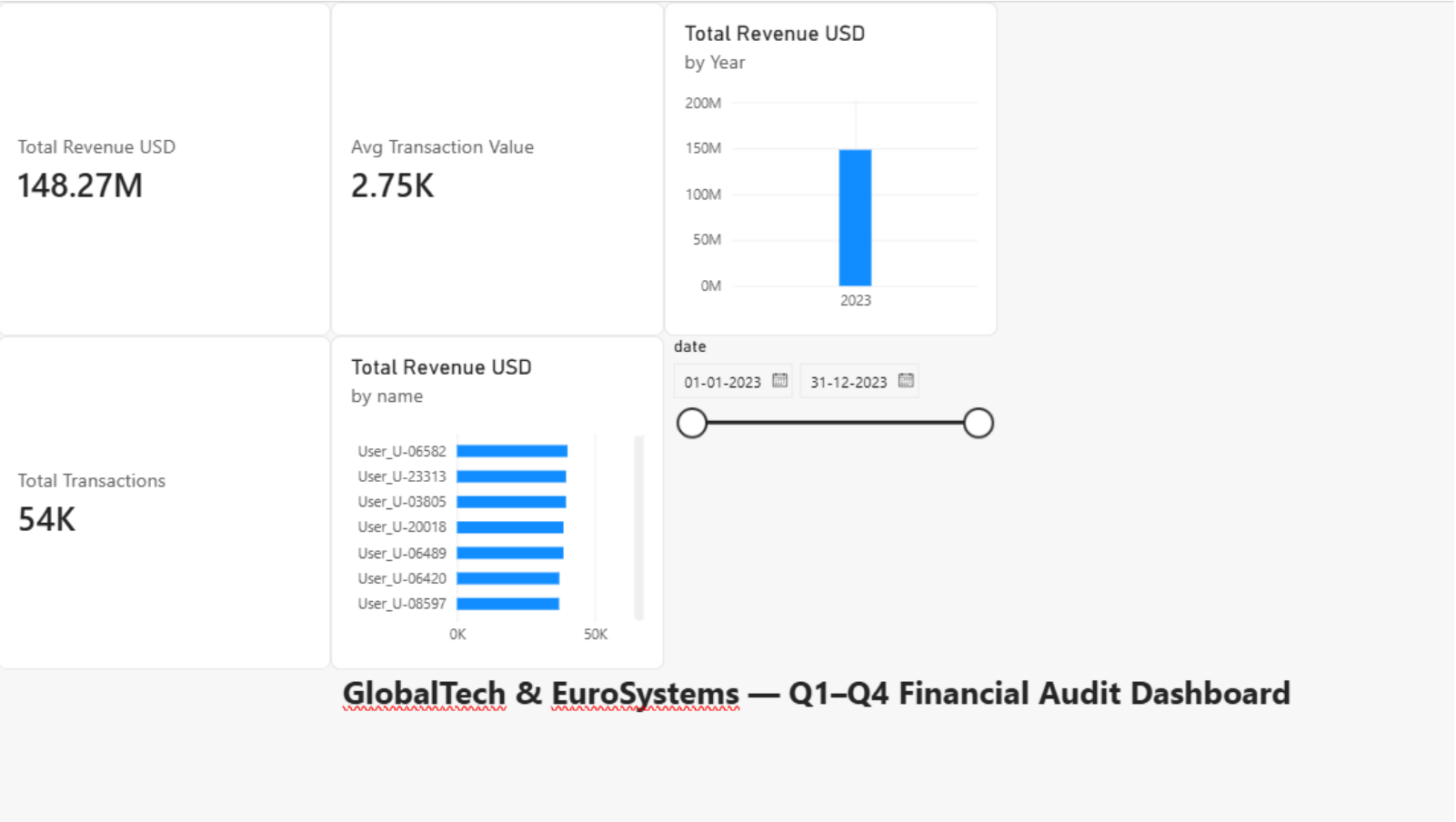
Top 5 Most Valuable Customers by Total USD Revenue



2023 Monthly USD Revenue Trend



Results



Business Impact & Operational ROI

- Explain the business value of building this automated pipeline rather than relying on manual spreadsheets.
- Time Efficiency: Reduces financial reconciliation processing time from days of manual Excel work to under 2 seconds.
- Error Reduction: Completely removes human error in manual currency conversions and log filtering.
- Scalability: The pipeline can easily handle millions of rows as GlobalTech scales internationally without requiring additional headcount.

Future Scope & Technical Enhancement Roadmap

- 4-step vertical roadmap timeline.
- 1. Pipeline Orchestration & Scheduling: Transition manual execution to automated AWS Lambda or Apache Airflow DAGs for nightly ETL updates.
- 2. Live Exchange Rate API Feeds: Connect directly to the European Central Bank API for real-time currency conversion rates.
- 3. Real-Time Log Ingestion: Implement Apache Kafka message streaming to filter error logs at ingestion time.
- 4. Predictive Financial Analytics: Integrate machine learning time-series models (Prophet/ARIMA) in Python to forecast quarterly revenue performance.

Strategic Recommendations

- 3 action-oriented callout blocks.
- 1. Enterprise Customer Retention: Focus key account management efforts on high-value enterprise accounts driving recurring volume.
- 2. Productionizing the Dashboard: Deploy GlobalTech Financial_Audit_Dashboard.pbix to Power BI Service for scheduled cloud refreshes.
- 3. Automated Log Filtering: Require engineering teams to separate error logs into dedicated log files at source.

Conclusion

- **Automated Data Reconciliation:** Successfully engineered an end-to-end Python ETL pipeline that ingests, cleans, deduplicates, and merges multi-source enterprise data in under 2 seconds.
- **Audited Single Source of Truth:** Reconciled 53,899 active transactions across 2023, establishing an audited total global revenue of \$148,274,622.91 USD.
- **Data Quality & Risk Mitigation:** Removed system error logs via vectorized Regex string matching, deduplicated account status histories through SQL window functions, and eliminated unrecorded weekend transactions via temporal forward-filling.
- **Executive Visibility:** Delivered an executive Power BI audit dashboard (GlobalTech Financial Data Analytics & Audit.pbix) offering dynamic slicing across quarterly performance and top enterprise customer contributions

Project links

- ▶ GitHub Link:

<https://github.com/ayushkumar1311/GlobalTech-Financial-Data-Analytics-Audit.git>

- ▶ Google Collob:

https://colab.research.google.com/drive/128pGzLybn_OGt-UiZCpFZ136gcSmyK7r?usp=sharing